GGG-T-0070C(GSA-FSS)
October 18, 1968
INTERIM REVISION OF
Fed. Spec. GGG-T-70B
March 14, 1961

INTERIM FEDERAL SPECIFICATION

TAP, THREAD CUTTING (HIGH SPEED STEEL, GROUND THREAD, HAND AND SPARK PLUG)

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D.C. 20405, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification GGG-T-70B, dated March 14, 1961.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers high speed steel ground thread hand taps in machine screw and fractional sizes and metric spark plug taps. These taps are used for machine and hand tapping.

1.1.1 Federal specification coverage. This specification does not include all types of taps indicated by the title of this specification, or which are commercially available, but is intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and series. The taps herein shall be of the following types, series, thread limits, and styles as specified, (see 6.1):

Type I. Machine screw and fractional sizes, standard point.
- Series A. Coarse thread (NC, UNC).
- Series B. Fine thread (NF, UNF).
- Series C. Special threads (NS).
- Series D, E, F and G. See 6.2
  - Style a. Taper
  - Style b. Plug.
  - Style c. Bottoming.

Type II. Machine screw and fractional sizes, spiral point.
- Series A. Coarse thread (NC, UNC).
- Series B. Fine thread (NF, UNF).
  - Style b. Plug.
  - Style c. Bottoming.

Type III. Spark plug (metric).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification.

Federal Specification:

QQ-T-590 - Tool Steel, High-Speed.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424 - Hand Tools, Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. The issue in effect on date of invitation for bids shall apply.

United States of America Standards Institute (USASI) Standards:

B94.9 - 1967 Taps, Cut and Ground Threads.
Z76.4 - 1967 Standard Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superificial Hardness, and Knoop Hardness) (ASTM E140-67).

3 REQUIREMENTS

3.1 Design and construction. Taps shall be designed and constructed to meet the requirements of B94.9 for high speed steel, ground thread, regular hand or regular machine screw taps as specified, except type III taps shall have a metric form thread with dimensions as shown herein. All taps to and including 3/8 inch size shall have external centers on thread and shank ends; the center on thread end shall be removed on bottoming taps. All taps larger than 3/8 inch shall have internal centers in thread and shank ends.

Style A. Taper taps shall have the first 8 to 10 threads chamfered.
Style B. Plug taps shall have the first 3 to 5 threads chamfered.
Style C. Bottoming taps shall have the first 1 to 1-1/2 threads chamfered.

Unless otherwise specified (see 6.1), right-hand thread-cutting taps shall be furnished.

3.2 Material. The taps shall be made from high speed tool steel equal in strength and durability to that conforming to QQ-T-590.

3.3 Finish. All surfaces of the taps shall be ground in accordance with good commercial practice.

3.4 Hardness. Taps shall be through hardened all over to the following Rockwell "C" hardness (see 4.4.2): Tap sizes number 6 and smaller -- RC 62-65, tap sizes larger than number 6 -- RC 63-66. Shanks may be drawn tougher in accordance with the manufacturer's standard practice.

3.5 Performance. There shall be no evidence of chipping, distortion, or wear when tested in accordance with 4.4.1.

3.6 Cases. When specified (see 6.1), tap sets shall be furnished in wood, metal, or plastic cases. The cases shall contain permanent separators suitable for keeping the taps from contacting each other. The filled cases shall resist a drop of 3 feet to a concrete floor without fracture.

3.7 Type I. Type I taps shall be the regular hand tap design with standard points, and of the size, thread series, style and pitch limit specified (see 6.1 and table I).

3.8 Type II. Type II taps shall be the regular hand tap design with spiral point and shallow flutes, and of the size, thread series, style, and pitch limit specified (see 6.1 and table II).

3.9 Type III. Type III taps shall be suitable for cutting threads for 14 and 18mm spark plugs and conform to tables III and IV for the size specified (see 6.1).

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TABLE I. Type I, standard point taps

Size	Threads per inch			Number of flutes		Pitch diameter limits and styles*						
	NC UNC	NF UNF	NS	STD.	OPTIONAL	H-1	H-2	H-3	H-4	H-5	H-6	H-7
0	---	80	--	2	---	TPB	PB	---	---	---	---	---
1	64	---	--	2	---	TPB	P	---	---	---	---	---
1	---	72	--	2	---	TPB	PB	---	---	---	---	---
2	56	---	--	3	---	TPB	TPB	---	---	---	---	---
2	56	---	--	--	2	PB	PB	---	---	---	---	---
2	---	64	--	3	---	P	TPB	---	---	---	---	---
3	48	---	--	3	---	P	TPB	---	---	---	---	---
3	48	---	--	--	2	---	PB	---	---	---	---	---
3	---	56	--	3	---	P	TPB	---	---	---	---	---
4	40	---	--	3	---	TPB	TPB	---	---	---	---	---
4	40	---	--	--	2	P	PB	---	---	---	---	---
4	---	48	--	3	---	P	TPB	---	---	---	---	---
4	---	---	36	3	---	---	TPB	---	---	---	---	---
5	40	---	--	3	---	TPB	TPB	---	---	---	---	---
5	40	---	--	--	2	---	PB	---	---	---	---	---
5	---	44	--	3	---	---	TPB	---	---	---	---	---
5	---	44	--	--	2	---	P	---	---	---	---	---
6	32	---	--	3	---	TPB	TPB	TPB	---	---	---	PB
6	32	---	--	--	2	P	PB	PB	---	---	---	---
6	---	40	--	3	---	P	TPB	---	---	---	---	---
6	---	40	--	--	2	---	P	---	---	---	---	---
8	32	---	--	4	---	TPB	TPB	TPB	---	---	---	PB
8	32	---	--	--	2	P	PB	PB	---	---	---	---
8	32	---	--	--	3	PB	PB	PB	---	---	---	PB
8	---	36	--	4	---	P	TPB	---	---	---	---	---
10	24	---	--	4	---	TPB	TPB	TPB	---	---	---	PB
10	24	---	--	--	2	---	PB	PB	---	---	---	---
10	24	---	--	--	3	P	P	PB	---	---	---	PB
10	---	32	--	4	---	TPB	TPB	TPB	---	---	---	PB
10	---	32	--	--	2	PB	PB	PB	---	---	---	---
10	---	32	--	--	3	P	PB	PB	---	---	---	PB
12	24	---	--	4	---	P	---	TPB	---	---	---	---
12	---	28	--	4	---	P	---	TPB	---	---	---	---
1/4	20	---	--	--	2	---	---	PB	---	---	---	---
1/4	20	---	--	--	3	P	P	PB	---	PB	---	---
1/4	20	---	--	4	---	TPB	TPB	TPB	---	PB	---	---
1/4	---	28	--	--	2	---	---	PB	---	---	---	---
1/4	---	28	--	--	3	---	---	PB	---	---	---	---
1/4	---	28	--	4	---	PB	PB	TPB	PB	---	---	---
5/16	18	---	--	--	2	---	---	PB	---	---	---	---
5/16	18	---	--	---	3	P	P	PB	---	PB	---	---
5/16	18	---	--	4	---	PB	TPB	TPB	---	PB	---	---
5/16	---	24	--	--	3	---	---	PB	---	---	---	---
5/16	---	24	--	4	---	PB	PB	TPB	PB	---	---	---
3/8	16	---	--	--	3	P	---	PB	---	PB	---	---
3/8	16	---	--	4	---	PB	TPB	TPB	---	PB	---	---
3/8	---	24	--	--	3	---	---	PB	---	---	---	---
3/8	---	24	--	4	---	PB	PB	TPB	PB	---	---	---
7/16	14	---	--	--	3	---	---	PB	---	---	---	---
7/16	14	---	--	4	---	PB	PB	TPB	---	PB	---	---
7/16	---	20	--	--	3	---	---	P	---	---	---	---
7/16	---	20	--	4	---	PB	PB	TPB	---	PB	---	---

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TABLE I. Type I, Standard point taps (con.)

Size	Threads per inch			Number of flutes		Pitch diameter limits and styles*						
	NC UNC	NF UNF	NS	STD.	OPTIONAL	H-1	H-2	H-3	H-4	H-5	H-6	H-7
1/2	13	—	—	—	3	—	—	PB	—	—	—	—
1/2	13	—	—	4	—	PB	PB	TPB	—	PB	—	—
1/2	—	20	—	—	3	—	—	P	—	—	—	—
1/2	—	20	—	4	—	PB	PB	TPB	—	PB	—	—
9/16	12	—	—	4	—	—	P	TPB	—	PB	—	—
9/16	—	18	—	4	—	—	P	TPB	—	PB	—	—
5/8	11	—	—	4	—	P	P	TPB	—	PB	—	—
5/8	—	18	—	4	—	P	P	TPB	—	PB	—	—
11/16	—	—	11,16	4	—	—	—	TPB	—	—	—	—
3/4	10	—	—	4	—	P	P	TPB	—	PB	—	—
3/4	—	16	—	4	—	P	P	TPB	—	PB	—	—
7/8	9	—	—	4	—	—	P	—	TPB	—	P	—
7/8	—	14	—	4	—	—	P	—	TPB	—	P	—
1	8	—	—	4	—	—	P	—	TPB	—	P	—
1	—	12	—	4	—	—	—	—	TPB	—	—	—
1	—	—	14	4	—	—	P	—	TPB	—	—	—
1-1/8	7	12	—	4	—	—	—	—	TPB	—	—	—
1-1/4	7	—	—	4	—	—	—	—	TPB	—	—	—
1-1/4	—	12	—	6	—	—	—	—	TPB	—	—	—
1-3/8	6	—	—	4	—	—	—	—	TPB	—	—	—
1-3/8	—	12	—	6	—	—	—	—	TPB	—	—	—
1-1/2	6	—	—	4	—	—	—	—	TPB	—	—	—
1-1/2	—	12	—	6	—	—	—	—	TPB	—	—	—

*T = taper, P = plug, B = bottoming

TABLE II. Type II, spiral point taps

Size	Threads per inch			Number of flutes		Pitch diameter limits and styles*					
	NC UNC	NF UNF	NS	STD.	OPTIONAL	H-1	H-2	H-3	H-4	H-5	H-7
0	—	80	—	2	—	PB	PB	—	—	—	—
1	64	—	—	2	—	P	P	—	—	—	—
1	—	72	—	2	—	P	P	—	—	—	—
2	56	—	—	2	—	PB	PB	—	—	—	—
2	—	64	—	2	—	P	PB	—	—	—	—
3	48	—	—	2	—	P	PB	—	—	—	—
3	—	56	—	2	—	P	P	—	—	—	—
4	—	—	36	2	—	—	P	—	—	—	—
4	40	—	—	2	—	PB	PB	—	—	—	—
4	—	48	—	2	—	P	PB	—	—	—	—
5	40	—	—	2	—	PB	PB	—	—	—	—
5	—	44	—	2	—	—	P	—	—	—	—
6	32	—	—	2	—	PB	PB	PB	—	—	PB
6	—	40	—	2	—	P	PB	—	—	—	—
8	32	—	—	2	—	PB	PB	PB	—	—	PB
8	—	36	—	2	—	P	P	—	—	—	—
10	24	—	—	2	—	P	PB	PB	—	—	P

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TABLE II. Type II, spiral point taps (con.)

Size	Threads per inch			Number of flutes		Pitch diameter limits and styles*					
	NC UNC	NF UNF	NS	STD.	OPTIONAL	H-1	H-2	H-3	H-4	H-5	H-7
10	—	32	—	2	—	PB	PB	PB	—	P	P
12	24	—	—	2	—	P	—	PB	—	—	—
12	—	28	—	2	—	—	—	P	—	—	—
1/4	20	—	—	2	—	P	P	PB	—	P	—
1/4	20	—	—	—	3	—	—	P	—	P	—
1/4	—	28	—	2	—	P	P	PB	P	—	—
1/4	—	28	—	—	3	—	P	—	P	—	—
5/16	18	—	—	2	—	P	P	PB	—	P	—
5/16	18	—	—	—	3	—	—	P	—	P	—
5/16	—	24	—	2	—	P	P	PB	P	—	—
5/16	—	24	—	—	3	—	P	—	P	—	—
3/8	16	—	—	3	—	P	P	P	—	P	—
3/8	—	24	—	3	—	P	P	P	P	—	—
7/16	14	—	—	3	—	—	P	P	—	P	—
7/16	—	20	—	3	—	—	P	P	—	P	—
1/2	13	—	—	3	—	P	P	P	—	P	—
1/2	—	20	—	3	—	P	P	P	—	P	—
5/8	11	—	—	3	—	—	—	P	—	P	—
3/4	10	—	—	3	—	—	—	P	—	P	—

*P = plug, B = bottoming.

TABLE III. General dimensions for type III, spark plug (metric), taps

Nom. size	Pitch	Thread length ± 3/32	Overall length ± 1/32	No. of flutes	Size of square -0.006	Length of square ± 1/32	Diameter of shank
14mm	1.25mm	1-21/32 inch	3-19/32 inch	4	0.322	1/2	0.429/0.4275
18mm	1.50mm	1-13/16 inch	4-1/32 inch	4	.406	5/8	.542/ .540

TABLE IV. Thread limits for type III, spark plug (metric), taps

Nom. size	Major diameter (Inch)			Pitch diameter (Inch)		
	Basic	Min.	Max.	Basic	Min.	Max.
14mm	0.5512	0.5550	0.5560	0.5192	0.5207	0.5212
18mm	.7087	.7135	.7145	.6703	.6718	.6723

3.10 Workmanship. Workmanship shall be first class in all respects and equal to good commercial practice. Taps shall be clean, free from pits, cracks, corrosion, and other defects.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, and in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table V.

TABLE V. Sampling data

Category	Sample unit	Inspection level	Acceptable Quality Level	AQL expressed in terms of	Reference
Visual examination	One each	II	1.5	Percent defective	4.3.1
Dimensional examination	One each	II	1.5	Percent defective	4.3.2
Testing	One each	S.4	1.5	Percent defective	4.4
Preparation for delivery	One each	II	4.0	Defects per hundred units	4.5

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, construction, workmanship, and marking. Defects are listed in table VI.

TABLE VI. Defects list

Examine	Defect
Design	Type, series, style, and size not as specified.
Finish	Not as specified.
Construction	Not as specified.
Workmanship	Not as specified.
Marking	Not as specified.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 and 4.4.2.

4.4.1 Performance test. Select a tap drill which will allow approximately 75 percent of thread in the hole to be tapped. Drill, then tap ten holes in a mild-steel plate having an approximate thickness equal to the outside diameter of the tap. Use an approved type lubricant while tapping holes. In tapping with spiral pointed taps, the inspector shall note that the action of the tap is such that the chips are curled out ahead of the tap. Check first and tenth hole tapped with a "Go" and "Not go" gage to check uniformity of hole produced. If any difference exists between the first and tenth hole, recheck tap used for possible dullness, breakage at cutting edge or other failure.

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4.4.2 Hardness test. Hardness determination shall be made with appropriate hardness testing equipment with the readings converted, if necessary, to Rockwell "C" scale in accordance with USASI Z76.4-1967 (ASTM E-140-1967).

5. PREPARATION FOR DELIVERY

5.1 Cleaning, preservation, packaging, and packing. Cleaning, preservation, packaging, and packing shall be as specified in MIL-H-15424. The level of cleaning, preservation, and packing shall be level A or C, as specified (see 6.1), and the level of packing level A, B, or C, as specified (see 6.1).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special markings required by the contract or order (see 6.1), interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. Marking shall be as specified in MIL-H-15424.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(c) Type, series, thread limits, style and size required (see 1.2.1, and tables I and II).
(d) If case is required for tap sets (see 3.6).
(e) Special markings required (see 5.2.1).

6.2 Deletion data. Series D, E, F, and G taps formerly covered by this specification have been deleted for lack of demand.

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